Notes on exponentially decaying viscosity

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The surface stress angle is defined as

$$\theta = \arg\left[F(z=0)\right] = \arg\left[A_1 + B_1\right]. \tag{1}$$

Summing A_1 and B_1 gives

$$A_1 + B_1 = \nu_1 U_g \frac{1+i}{h_{\text{Ek1}}} \cdot \frac{-(1-\gamma) + e^{2(1+i)\varphi}(1+\gamma)}{1-\gamma + e^{2(1+i)\varphi}(1+\gamma)}. \tag{2}$$

To simplify this expression, one can factor out $(1+\gamma)$ from both the numerator and denominator, yielding

$$A_1 + B_1 = \nu_1 U_g \frac{1+i}{h_{\text{Ek1}}} \cdot \frac{e^{2(1+i)\varphi} - \frac{1-\gamma}{1+\gamma}}{\frac{1-\gamma}{1+\gamma} + e^{2(1+i)\varphi}} = \nu_1 U_g \frac{1+i}{h_{\text{Ek1}}} \cdot \frac{e^{2(1+i)\varphi} - R}{R + e^{2(1+i)\varphi}}. \tag{3}$$

Dividing the numerator and denominator by $e^{(1+i)\varphi}$ one obtains

$$A_1 + B_1 = \nu_1 U_g \frac{1+i}{h_{\text{Ek1}}} \cdot \frac{e^{(1+i)\varphi} - Re^{-(1+i)\varphi}}{e^{(1+i)\varphi} + Re^{-(1+i)\varphi}}. \tag{4}$$

Since ν_1, U_g and h_{Ek1} are all real and positive, their argument is zero. The factor $(1+i)$ contributes $\arg[1+i] = \pi/4$. The surface stress angle therefore becomes

$$\theta = \frac{\pi}{4} + \arg\left[\frac{e^{(1+i)\varphi} - Re^{-(1+i)\varphi}}{e^{(1+i)\varphi} + Re^{-(1+i)\varphi}}\right]. \tag{5}$$

In the single-layer limit $\gamma \to 1$, one has $R \to 0$ and the argument of the fraction vanishes, recovering the classical Ekman result $\theta = \pi/4 = 45^\circ$.